



SVP-25-061

10 CFR 50.73

November 4, 2025

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555

Quad Cities Nuclear Power Station, Unit 1
Renewed Facility Operating License No. DPR-29
NRC Docket No. 50-254

Subject: Licensee Event Report 254/2025-006-00 "High Pressure Coolant Injection Inoperable Due to Turbine Inlet Valve Failure to Open"

Enclosed is Licensee Event Report 254/2025-006-00 "High Pressure Coolant Injection Inoperable Due to Turbine Inlet Valve Failure to Open".

This report is being submitted in accordance with 10 CFR 50.73(a)(2)(v)(D) for an event or condition that could have prevented the fulfillment of a safety system needed to mitigate the consequences of an accident.

There are no regulatory commitments contained in this letter.

Should you have any questions concerning this report, please contact Conner Bealer at 779-231-6207.

Respectfully,

A handwritten signature in black ink, appearing to read "Doug Hild", written over a white background.

Doug Hild
Site Vice President
Quad Cities Nuclear Power Station

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – Quad Cities Nuclear Power Station



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Quad Cities Nuclear Power Station, Unit 1

☒ 050
☐ 052

2. Docket Number

00254

3. Page

1 OF 4

4. Title

High Pressure Coolant Injection Inoperable Due to Turbine Inlet Valve Failure to Open

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
09	05	2025	2025	- 006 -	00	11	04	2025	N/A	☐ 050	N/A
									N/A	☐ 052	N/A

9. Operating Mode	10. Power Level
1	100%

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☐ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☐ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☐ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Richard Swart – Regulatory Assurance	309-227-2810

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
A	BJ	BKR	C770	Y	n/a				

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)	15. Expected Submission Date	Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On September 4, 2025 at 1845 during the performance of a Unit 1 High Pressure Coolant Injection (HPCI) Operability Test, it was identified that the system's steam inlet motor operated valve (MOV) failed to open. The inability to open this valve prevented the steam driven system from fulfilling its function of providing emergency core cooling. The cause of this condition, discovered on September 5, 2025, was a degraded breaker which was found to contain a degraded open coil and a binding contact roller. This prevented electrical power from being provided to the valve. The degraded coil was replaced and the roller attachment was cleaned, lubricated, and reinstalled.

The breaker failed due to an inadequate maintenance strategy.

This report is being submitted in accordance with 10 CFR 50.73(a)(2)(v)(D) for an event or condition that could have prevented the fulfillment of a safety function needed to mitigate the consequences of an accident.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME Quad Cities Nuclear Power Station Unit 1	☒ 050	2. DOCKET NUMBER 00254	3. LER NUMBER		
	☐ 052		YEAR 2025	SEQUENTIAL NUMBER - 006	REV NO. - 00

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

General Electric – Boiling Water Reactor, 2957 Megawatts Thermal Rated Core Power

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].

EVENT IDENTIFICATION

High Pressure Coolant Injection Inoperable Due to Turbine Inlet Valve Failure to Open

A. CONDITION PRIOR TO EVENT

Unit: 1	Event Date: September 5, 2025	Event Time: 1845 hours CST
Reactor Mode: 1	Mode Name: Power Operation	Power Level: 100%

No systems, structures, or components that were inoperable at the start of the event contributed to the event.

B. DESCRIPTION OF EVENT

On September 4, 2025 at 1845 during the performance of a High Pressure Coolant Injection (HPCI) [BJ] Operability Test, operators discovered that the Turbine Inlet Motor Operated Valve (MOV) [V] 1-2301-3 would not open. Multiple attempts were made to open the MOV from the control switch and the associated breaker contacts could not be heard cycling.

Troubleshooting on September 5, 2025 at 0130, determined that the electrical breaker [BKR] feeding the MOV had an opening coil [CL] that had failed, and rollers associated with the coil assembly were binding. Discovery of the opening coil degradation confirmed that the MOV would fail to open in either manual or automatic HPCI demands, as opposed to failure to operate only from the valve control switch. The coil was replaced, and the roller assembly was cleaned and relubricated. Post maintenance testing of the MOV was completed on September 5, 2025 at 2228. The MOV breaker prevented proper HPCI operation for 27 hours and 43 minutes.

The HPCI Operability Test was subsequently completed with satisfactory results.

Event Notification System (ENS) report number 57905 was submitted to the NRC on September 5, 2025.

This report is being submitted in accordance with 10 CFR 50.73(a)(2)(v)(D) for an event or condition that could have prevented the fulfillment of a safety function needed to mitigate the consequences of an accident.

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NARRATIVE**C. CAUSE OF EVENT**

The self-revealing cause of the event was an inadequate maintenance strategy applied to the associated breaker which allowed contactor lubrication to degrade such that it prevented contacts from closing. A preventative maintenance task was retired which inspected the associated breaker while also cleaning and lubricating the contactors.

D. SAFETY ANALYSISSystem Design

The HPCI system is designed to ensure that adequate core cooling takes place for all break sizes less than those for which Low Pressure Coolant Injection (LPCI) or Core Spray systems can adequately protect the core without assistance from other engineered safety features.

The purpose of the breaker in question is to permit power to the Unit 1 HPCI Turbine Inlet MOV during both automatic and manual system initiation. Without the ability to power and open this valve, steam is unable to reach the HPCI turbine. Without this turbine driving the HPCI pumps, the system is not capable of injection.

Safety Impact

The safety significance of this event is minimal. At the time of the event, Unit 1 was at full power in a normal electric plant lineup with all other means of Reactor Coolant Injection available.

The Unit 1 Reactor Core Isolation Cooling (RCIC) system and the Unit 1 Automatic Depressurization System (ADS) remained Operable throughout the event. There was no impact on the health and safety of the public or plant personnel.

This event was within the analysis of UFSAR Chapter 15. There was no radioactive release associated with this event.

This event is considered a Safety System Functional Failure per NEI 99-02, Revision 8.

E. CORRECTIVE ACTIONSImmediate:

1. The breaker coil was replaced. The roller assembly was cleaned and relubricated.

Follow up:

1. Revise the breaker's maintenance strategy.
2. Conduct Strategic Engineering teaching and learning related to this event.



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NARRATIVE

F. PREVIOUS OCCURENCES

The station events database, LERs and INPO Industry Reporting Information System (IRIS) were reviewed for similar events at Quad Cities Nuclear Power Station. This event was caused by equipment failures causing operations to isolate the HPCI system. No previous occurrences in the previous three years were identified on this breaker.

G. COMPONENT FAILURE DATA

Thermal Overload Relay
Manufacturer: Eaton / QualTech NP
Nomenclature: Coil, Overload
Model/Part Number: H1029

This event has been reported to IRIS.